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New palaeomagnetic results from the Whin Sill complex: evidence for a multiple intrusion event and revised virtual geomagnetic poles for the late Carboniferous for the British Isles

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Abstract: A large-scale palaeomagnetic study (125 specimens from 88 sampling localities) has been carried out on the Whin Sill complex and its associated dykes in northern England. The dominant carrier of remanent magnetization of the doleritic rocks is relatively pure magnetite of pseudo-single domain size, and the characteristic remanent magnetizations are typically well defined and unidirectional. Regionally consistent palaeomagnetic directions are obtained from three geographically distinct parts of the Whin Sill complex, here named as the Holy Island Sill (Declination/Inclination $(D/I) = 189.5/-2.8$, $\alpha_{95} = 6.3$), the Alnwick Sill ($D/I = 194.6/-25.6$, $\alpha_{95} = 8.1$) and the Hadrian's Wall–Pennines Sill ($D/I = 189.2/3.3$, $\alpha_{95} = 3.5$). Although the Holy Island Sill and the Hadrian's Wall–Pennines Sill are of similar age (*c*. 296 Ma) and indistinguishable on palaeomagnetic grounds, the difference in palaeomagnetic directions between them and the Alnwick Sill demonstrates the multi-component nature of the complex; magma-flow indicators suggest that the Holy Island Sill and the Hadrian's Wall–Pennines Sill may also represent separate intrusion events. Dykes local to the three sills have remanence directions corresponding to their respective sills and can tentatively be identified as their feeder dykes. Virtual geomagnetic poles for the three palaeomagnetic sites are: Holy Island Sill and Holy Island Dyke, 346.8E, 35.4S; Alnwick Sill, High Green Dyke and St. Oswald's Chapel Dyke, 337.1E, 47.1S; Hadrian's Wall–Pennines Sill and Hett Dyke, 347.1E, 32.9S.

Keywords: sills, dykes, palaeomagnetism, emplacement, dolerite.

The late Carboniferous doleritic Whin Sill complex of northern England represents the archetype for sheet-like concordant intrusions; from it, indeed, geology acquired the term 'sill' (e.g. Dunham & Strasser-King 1982; Johnson & Dunham 2001). There is widespread evidence of igneous activity in NW Europe in the late Carboniferous and early Permian; for example, in the Midland Valley of Scotland (Dunham & Strasser-King 1982), the Oslo Rift (Neumann 1994; Sundvoll & Larsen 1994) and southern Sweden (Obst 1999). The suggestion of Ernst & Buchan (1997, 2001) that all these occurrences are related and are manifestations of a plume-related 'Jutland event', has refocused interest on igneous activity in the region. Although the Whin Sill complex has been studied for more than a century (references given by Dunham & Strasser-King 1982; Randall 1995; Johnson & Dunham 2001; and a Presidential Address by Francis (1982) to the Society), problems regarding its mode of emplacement and the location of feeder dykes still remain.

As the dolerites of the Whin Sill complex carry a stable characteristic remanent magnetization, palaeomagnetic results from the complex have provided an important datum. A study by Storetvedt & Gidskehaug (1969), following pioneering work by Creer *et al*. (1959), has provided a magnetization direction that is accepted as defining a key virtual geomagnetic pole (VGP) for the late Carboniferous, contributing to the apparent polar wander path of the British Isles (Torsvik *et al*. 1989; Piper *et al*. 1991).

This paper reports on an extensive palaeomagnetic study carried out on the Whin Sill complex and its associated dykes. The results of the study provide new information on both the intrusion history of the Whin Sill complex and its accepted VGP, revealing previously unresolved detail in both.

Geological setting and sampling

The Whin Sill complex is a large tholeiitic quartz dolerite intrusion in northern England (Fig. 1). On land its outcrop extends for *c*. 120 km in a north–south direction and *c*. 80 km in an east–west direction but, as the sill complex generally dips underneath the North Sea, its true extent is unknown. The complex has a maximum observed thickness of *c*. 80 m and its average thickness is of the order of 30 m. The stratigraphical horizons into which it is emplaced range from Dinantian to Westphalian and the overburden thickness at the time of emplacement has been estimated as between 0.5 and 1 km (Francis 1982).

Four major ENE–WSW-trending doleritic dykes are associated with the Whin Sill complex (Fig. 1); from north to south these are the Holy Island Dyke, the High Green Dyke, the St. Oswald's Chapel Dyke and the Hett Dyke. The thickest of these dykes is the High Green Dyke, which reaches a width of up to *c*. 65 m. The close geographical relationship and the petrological affinity between the dykes and the sill complex have led several workers to propose that the four dykes formed the feeders to the complex (Holmes & Harwood 1928; Anderson 1951; Francis 1982; Goulty *et al*. 2000). Additional geochemical data seem to confirm this finding (Thorpe & MacDonald 1985). Nevertheless, no direct relationship between the dykes and the Whin Sill complex can be demonstrated anywhere in the field, and there is some dispute in the literature as to whether the dykes acted as feeders (Smythe 1930; Johnson & Dunham 2001).

The Little Whin Sill, another dolerite intrusion, also occurs in the study area (Fig. 1). The Little Whin Sill forms a thin leaf above the main sill and is, in contrast to the main Whin Sill dolerite, olivine bearing (Dunham & Kaye 1965; Harrison 1968).

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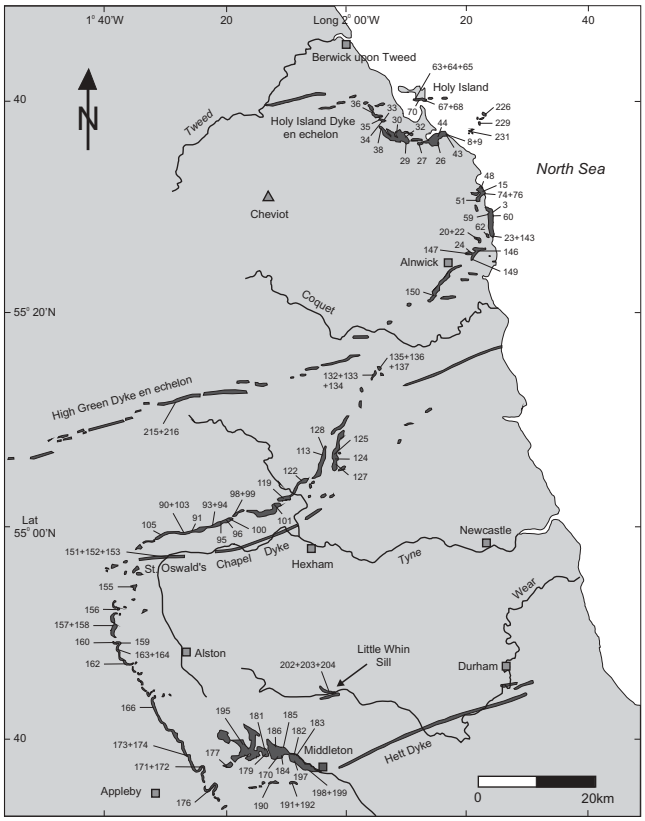


Fig. 1. Regional overview map of the Whin Sill complex and the associated dykes; the dolerite outcrops are plotted in black. Sample localities with corresponding specimen numbers are indicated.

The magma that fed the Little Whin Sill was less evolved and is believed to represent an early differentiate of the normal Whin Sill magma (Wadge *et al*. 1972; Johnson & Dunham 2001).

The age of the Whin Sill complex and the associated dykes can be narrowed down to Late Carboniferous to Early Permian by stratigraphical evidence. They intrude Upper Carboniferous host rocks (Westphalian B–C), but do not penetrate the Permian units (Dunham & Strasser-King 1982). However, clasts of Whin Sill dolerite are reported from breccias of Early Permian age from the Vale of Eden (Holmes & Harwood 1928; Dunham 1932). The age of the intrusion is also well constrained by several radiometric analyses. K–Ar analyses on 21 specimens suggest a mean age of 295 ± 6 Ma for the Whin Sill complex and the associated dykes (Fitch & Miller 1967) and a recent Ar/Ar study yielded an age of 294 ± 2 Ma for the Holy Island Dyke (Timmerman, pers. comm.). In addition, a provisional but precise U–Pb baddeleyite age of 297 Ma has been determined (Hamilton & Pearson, pers. comm.) from the southernmost part of the intrusion, in the area of Middleton (Fig. 1).

For this palaeomagnetic study, dolerite block samples from the Whin Sill complex and the four related dykes were collected from 88 localities (Fig. 1). The number of dyke outcrops sampled is relatively small as it is very difficult to find dyke exposures from which fresh, oriented specimens can be taken. The block samples were oriented in the field by magnetic compass; this procedure was checked and found adequate as no deflection of the compass needle, caused by the potentially high magnetization of the dolerite, could be observed.

Palaeodirectional analyses

The oriented dolerite block samples were drilled in the laboratory (Owens 1994). In total, 125 cylindrical samples of standard size (diameter 2.5 cm, length 2.2 cm) were analysed for their characteristic remanent magnetization (ChRM). The samples were stepwise thermally demagnetized in an MMTD1 furnace and their remanence was measured using an AGICO JR5a spinner magnetometer. Both furnace and magnetometer were constantly shielded against any external magnetic laboratory field.

In a pilot demagnetization study 24 samples were measured in steps of 50 °C from room temperature to 700 °C with two additional measurements at 525 and 575 °C. These analyses showed that most samples have a discrete unblocking temperature interval between 450 and 550 °C and that all samples are fully demagnetized at *c*. 575 °C. For routine demagnetization, measurements were therefore made at room temperature, followed by 50 °C intervals from 250 °C to a maximum temperature of 600 °C, with an additional measurement taken at 575 °C.

The ChRM of the individual samples was analysed using Zijderveld diagrams (Zijderveld 1967), and plots of normalized magnetic intensity (J/J_0) v. temperature were used to identify the related blocking temperature spectra. All samples revealed shallow southerly directed ChRM vectors that are generally very well defined (Fig. 2). Figure 3 shows four representative Zijderveld and intensity plots. All samples have a very strong remanent magnetization at room temperature (average $J_0 = 2566$ mA m^{-1}) and were fully demagnetized at *c*. 575 °C, which coincides with the Curie temperature of pure magnetite (e.g. Dunlop & Özdemir 2001). However, whereas the ChRM vectors for all samples are relatively consistent, the intensity v. temperature plots show distinctively different patterns and can be divided into four groups (Fig. 3). Most of the samples fall into group 1 (70%, Fig. 3a) and are characterized by a discrete blocking temperature interval of typically 50–150 °C with unblocking generally starting at 400–500 °C. This suggests that the magnetic mineralogy of these samples is dominated by magnetite and/or titanium-poor titanomagnetite (TM). Additionally, the well-defined blocking temperature interval indicates that the magnetic grain size of the samples is relatively homogeneous. Samples of group 2 (17%, Fig. 3b) show a distributed blocking temperature interval and the specimens typically lose their magnetization gradually until they are completely demagnetized at *c*. 575 °C. This demagnetization behaviour points either towards a heterogeneous magnetic grain-

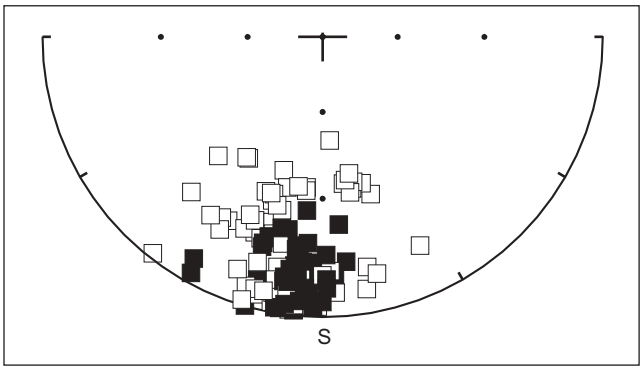


Fig. 2. Equal angle stereographic projection of the ChRM directions for all analysed specimens. ■, plots into the lower hemisphere of the stereonet; □, plots into the upper hemisphere.

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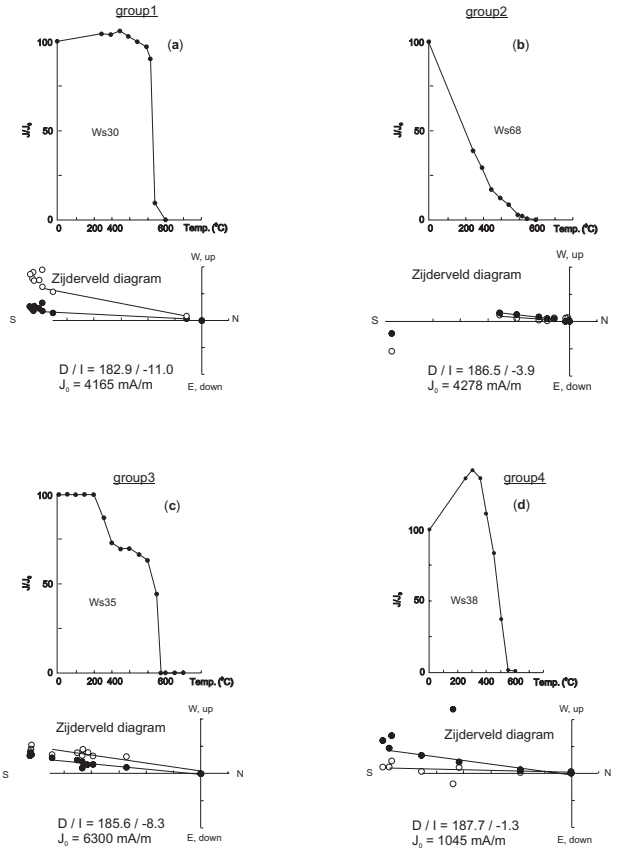


Fig. 3. Examples of typical thermal demagnetization behaviour of Whin Sill dolerite samples with their related Zijderveld diagrams. On the basis of the thermal demagnetization behaviour the samples can be divided into four groups: (a) group 1, 70% of all samples show a discrete unblocking interval; (b) group 2, 17% of the samples show a distributed unblocking interval; (c) group 3, 6% of the samples incorporate two magnetic carriers, pyrrhotite and magnetite; (d) group 4, 7% of the samples incorporate two magnetic vectors of opposing direction. The Zijderveld diagrams (orthogonal vector plots) illustrate the full component structure of remanence and the fitted line follows the trend of the isolated characteristic remanence direction. ●, projections onto the horizontal plane; ○, projections onto the vertical plane. *D*/*I* gives the value for the declination and inclination of the isolated ChRM; J_0 is the remanence intensity at room temperature.

size distribution with a wide range of unblocking temperature intervals or, alternatively, to a magnetic mineralogy in which the compositional range of the TMs is broader. The demagnetizing behaviour of group 3 samples (6%, Fig. 3c) is defined by two separate, discrete blocking intervals. The first unblocking occurs between 200 and 300 °C and may indicate the presence of monoclinic pyrrhotite (Curie temperature *c*. 300 °C, Dunlop & Özdemir 2001) whereas the second unblocking is again characteristic of magnetite and/or Ti-poor TM. Samples of group 4 (7%, Fig. 3d) are defined by a considerable initial increase of total magnetization up to 300 °C after which the specimens lose all their magnetization within a discrete unblocking interval. The initial increase in the total magnetization indicates that these samples incorporate magnetic vectors of relatively opposing directions. Stepwise thermal demagnetization gradually removes the effect of the softer, low-temperature constituents so that the magnetization carried by the magnetically harder constituents becomes dominant. No common ChRM direction could be isolated for any softer component. The temperature intervals at which the unblocking of the individual components occur indicate that the harder magnetic directions are carried by magnetite and/or Ti-poor TM whereas the softer magnetic directions are probably carried by Ti-rich TM, or possibly by monoclinic pyrrhotite.

Magnetic mineralogy and domain state

The reliability of palaeodirectional results is largely determined by the composition and domain state of the remanence carriers. In general, it is harder to overprint the direction of magnetization in single domain (SD) and pseudo-single domain (PSD) grains than it is in the larger multidomain (MD) grains. The former are therefore responsible for most of the stable remanence analysed in palaeomagnetic studies. To determine the composition and the magnetic grain size of the remanence carriers of the Whin Sill complex, the magnetic mineralogy has been studied microscopically and standard rock magnetic techniques have been applied to a representative subgroup of samples.

Reflection microscopy

Microscopic analyses reveal that the oxide mineralogy of the dolerites is dominated by magnetite and ilmenite. The two minerals typically occur as coarse-grained intergrowths and mostly comprise ilmenite as oxidation–exsolution lamellae in magnetite. The two minerals can also be present in a sandwich texture. Ilmenite is generally unaltered whereas magnetite shows varying degrees of alteration ranging from fairly fresh to heavily altered with a strongly pitted texture. Apart from the oxide phases smaller amounts of sulphides are present. Pyrrhotite is the most commonly occurring sulphide, sometimes being altered to hematite and magnetite along its basal cleavage; this suggests that the mineral occurs in its hexagonal variety, as monoclinic pyrrhotite does not possess this basal cleavage. Microprobe analyses on pyrrhotites in the dolerites of the Whin Sill complex have also suggested that the mineral is present in its hexagonal variety (Dunham & Strasser-King 1981). Whereas monoclinic pyrrhotite is ferrimagnetic, the hexagonal variety is antiferromagnetic (Dunlop & Özdemir 2001). This explains why, during stepwise demagnetization, significant magnetization loss was rarely recorded at the Curie temperature of pyrrhotite (group 3 behaviour; Fig. 3c), even though the mineral is present in virtually every sample.

Samples from within the first 10 cm from the sill contact are generally atypical in their opaque mineralogy. These specimens comprise ilmenite in a very fine skeletal texture and no magnetite is apparent.

Temperature-dependent susceptibility measurements

Low-field thermomagnetic curves (k–T curves) were measured for 12 samples using an AGICO KLY3-CS3 susceptibility bridge. For these analyses, the susceptibilities of rock powder samples were repeatedly measured as the samples were slowly heated to >700 °C and then cooled to room temperature. The analyses were performed under an argon atmosphere to prevent oxidation of the specimen's mineralogy by atmospheric oxygen during heating.

For the individual samples the maximum unblocking temperatures were determined graphically from the k–T heating curves. All samples are characterized by a sharp drop in susceptibility between *c*. 500 and 600 °C and have Curie temperatures between

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560 and 580 °C (Fig. 4). This behaviour shows that relatively pure magnetite is the dominant magnetic phase of the samples.

Only one sample (Fig. 4a) shows nearly reversible behaviour during heating and cooling, indicating that the specimen did not suffer significant chemical changes during the analysis. For most of the other samples the susceptibilities measured during cooling are significantly lower than those measured during heating (Fig. 4b). For these specimens it is proposed that at least some of the magnetite oxidized to maghemite and the lower susceptibility hematite.

All samples show the Hopkinson effect (Hopkinson, 1889), in which the susceptibility initially increases with rise in temperature. Thermal enhancement of the susceptibility occurs because, above the blocking temperature but below the Curie temperature, a magnetic grain behaves superparamagnetically (Clark & Schmidt 1982). The width of the Hopkinson peak, which determines over what temperature range there is significant enhancement of the susceptibility, is controlled by the width of the blocking temperature spectrum (Dunlop 1974). The height and the width of the Hopkinson peak can therefore provide a rough indicator of the magnetic grain size. The gentle increase in susceptibility with rise in temperature observed for specimen WS 195Ar (Fig. 4a) might reflect the presence of larger MD magnetite grains. In contrast, sample WS 113 (Fig. 4b) shows a relatively narrow Hopkinson peak, probably indicating the presence of PSD or small MD magnetite grains.

Hysteresis and saturation isothermal remanent magnetization (SIRM) properties

Hysteresis loop shape has been widely used to characterize magnetic properties. Day *et al*. (1977), by cross-plotting parameterized measures of loop shape, have devised a plot in which grains of different size and magnetic character fall in different fields. They plotted M_{rs}/M_s (the ratio of saturation remanence to saturation magnetization) against H_{cr}/H_c (the ratio of the coercivity of remanence to the coercive force). Coercivity of remanence, the back-field required to reduce a saturation remanence to zero, cannot readily be determined from hysteresis loop measurements and requires separate back-field measurements. However, a related property, the coercivity of remanent hysteresis, H_{rh}, which is determinable from the hysteresis loop, has been shown to be an adequate estimator of the coercivity of remanence (Fabian & von Dobeneck 1997; Karloukovski 2000).

In this study a limited number (18) of back-field experiments have been carried out to confirm the relationship between H_{cr} and H_{rh} (Fig. 5a). On this basis, by estimating H_{cr} from H_{rh}, a Day plot has been made for all 41 samples for which hysteresis curves were measured. All measurements were carried out on a Molspin Nuovo vibrating sample magnetometer and the results were analysed using the software package Hystear (von Dobeneck 1996).

The 18 specimens for which back-field measurements were made have values of H_{cr} between 10 and 63 mT. As previous workers have found, values of H_{rh} slightly overestimate H_{cr} (Fabian & von Dobeneck 1997; Karloukovski 2000). In estimating H_{cr}* (the asterisk is introduced to denote an estimated quantity) from H_{rh} the relationship $H_{cr}{}^* = 0.9\,H_{rh}$ has been used (Fig. 5a); the scatter observable in Figure 5a is too small to affect conclusions drawn from the Day plot (Fig. 5b). From the Day plot it is clear that most samples plot within a narrow area in the PSD field. Only two samples, WS 51 and WS 74, plot in the MD field.

Summary of the rock magnetic properties

Optical and magnetic analyses have shown that the dominant carrier of remanent magnetization of the dolerite of the Whin Sill complex is relatively pure magnetite, probably resulting from deuteric oxidation of original TM. The magnetite typically has an overall magnetic grain size in the PSD state; this might reflect a spread of grain sizes from SD, through PSD, to fine MD. A previous study on the magnetic mineralogy of the Whin Sill complex (Thomas *et al*. 1995) came to a similar conclusion. In addition, pyrrhotite is present in most samples but mainly in its hexagonal, aniferromagnetic variety. Only 7% of the analysed

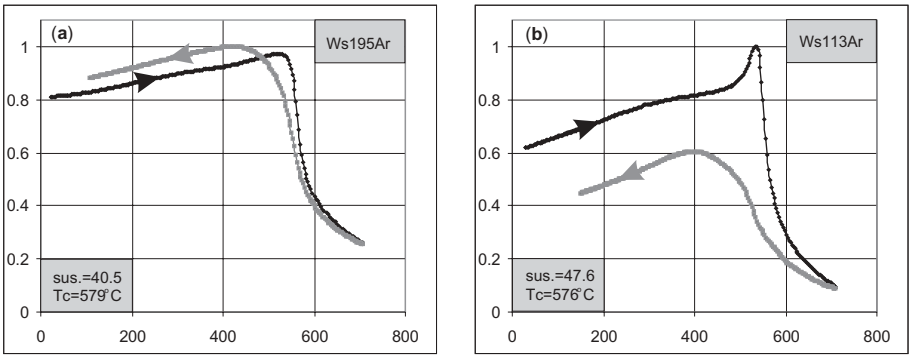


Fig. 4. Temperature v. normalized susceptibility k–T curves measured under an argon atmosphere. The heating and cooling curves are plotted in black and grey, respectively. (**a**) specimen WS 195; (**b**) specimen WS 113.

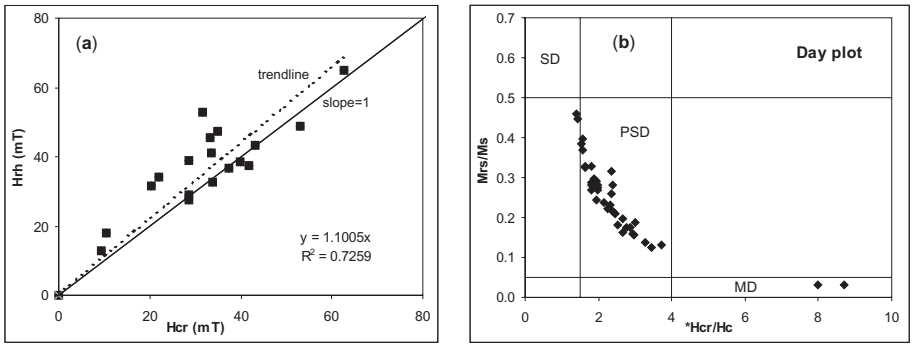


Fig. 5. (**a**) Correlation between H_{cr} and H_{rh}. The relationship is linear and H_{rh} is typically 10% higher than H_{cr}. (**b**) Modified Day plot for the Whin Sill samples. The individual grain size fields are defined as: SD ($M_{rs}/M_s > 0.5$ and $H_{cr}{}^*/H_c < 1.5$); PSD ($0.05 < M_{rs}/M_s < 0.5$ and $1.5 < H_{cr}{}^*/H_c < 4$); MD ($M_{rs}/M_s < 0.05$ and $H_{cr}{}^*/H_c > 4$).

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specimens contain monoclinic pyrrhotite, which can contribute to the total remanent magnetization of the rocks.

Hysteresis measurements showed that the two samples WS 51 and WS 74 are significantly softer magnetically than the typical samples. It is likely that at least some of the magnetite in these specimens has grown hydrothermally. That these samples also have the highest bulk susceptibilities ($sus_0 > 100 \times 10^{-3}$, Appendix 1) as expected for MD magnetite adds to this suggestion. The two samples were therefore omitted from the statistical palaeodirectional analyses.

The two specimens WS 8-211 and WS 8-233 were also rejected from further analysis as these specimens came from the contact zone (<10 cm) of the sill. Microscopy studies revealed that the opaque mineralogy of these samples is atypical; no magnetite was observed and skeletal ilmenite represents the dominant oxide phase. Further, the two specimens have susceptibilities of *c*. 0.85×10^{-3}, compared with an average value of 32.5×10^{-3} obtained from all specimens.

Palaeodirectional results

The remanence measurements show that the samples of the Whin Sill complex have a strong room-temperature magnetization (average 2566 mA m^{-1}, Appendix 1). In general, thermal demagnetization yielded only a single remanence component in a specimen (e.g. Fig. 3a). Only occasionally could a second, softer palaeomagnetic component be isolated. These softer components are removed at temperatures of 250 °C and are randomly oriented. The ChRM of all specimens analysed is southerly directed, of reversed polarity and well defined (Appendix 1). Block and site averages are listed in Appendix 2, together with statistical parameters.

It is not immediately clear whether a tilt correction should be applied to the dataset. In the study area the host rock bedding is typically gently dipping and dips rarely exceed 15°. There is evidence that the host rock stratigraphy was tilted at least to some degree, prior to the intrusion of the Whin Sill complex. In the northernmost part of the study area the complex is emplaced as a flat sheet into the Holburn anticline, cutting through the eastern fold limb into the core of the anticline (e.g. Robson 1977). This relationship clearly demonstrates that the area was deformed prior to the intrusion event. On this basis, it will be assumed here that no significant tilting of the Whin Sill complex occurred after the intrusive body acquired its magnetization; palaeodirectional results are quoted without any tilt correction applied.

Discussion

By plotting the palaeodirectional data for the individual regions of the Whin Sill complex and the associated dykes it becomes apparent that there are two distinctively different signatures (Fig. 6). For most of the Whin Sill complex the data are southerly directed with small positive or negative inclinations. A different pattern is observed for the Whin Sill complex in the Alnwick area, for the High Green Dyke and for the St. Oswald's Chapel Dyke. The data for these three regions are more steeply inclined and only of negative inclination. The mean palaeomagnetic directions for the individual intrusions are summarized in Table 1 and the detailed results for individual specimens and block averages are given in Appendix 1 and 2.

Field observations do not support significant post-emplacement tilting of the Whin Sill complex. The two different signatures are therefore unlikely to be due to tectonic block tilting or block rotation. Thus, the different palaeomagnetic patterns must reflect at least two igneous events during which local field inclination was distinctively different. This in turn means that, today, we are dealing with at least four geographically separated sill bodies (Fig. 6). Three of these are named, from north to south, as the Holy Island Sill (Fig. 6, 1), the Alnwick Sill (Fig. 6, 2), and the Hadrian's Wall–Pennines Sill (Fig. 6, 3). The Little Whin Sill (Fig. 6, 4) cannot be differentiated on the basis of the palaeomagnetic results from the Hadrian's Wall–Pennines Sill but differences in its petrology (Dunham & Kaye 1965; Harrison 1968) show that this smaller sill was emplaced prior to the other intrusions (Wadge *et al*. 1972; Johnson & Dunham 2001). Further discussion will focus only on the three larger sills.

The regionally plotted data also suggest relationships between the dykes and the individual sills. The Holy Island Dyke has a very similar palaeomagnetic signature to the Holy Island Sill (Fig. 6), which suggests that this dyke formed the feeder to the Holy Island Sill. The palaeomagnetic directions for the St. Oswald's Chapel Dyke and the High Green Dyke are within the error limits of the more steeply inclined direction of the Alnwick Sill, indicating that the two dykes and the sill might have been emplaced contemporaneously. The closer geographical location of the High Green Dyke to the Alnwick Sill suggests that this dyke may have fed the sill in the Alnwick area. In the southern part of the study area the Hett Dyke is the only dyke that has a similar palaeomagnetic signature to the Hadrian's Wall–Pennines Sill. It is therefore likely that the Hett Dyke formed the feeder to this sill. Dyke data are sparse and relationships between the individual dykes and sills cannot be unequivocally proven. Nevertheless, the combined evidence of related palaeomagnetic directions and of proximity argue for the proposed feeder relationships.

Further evidence that the Whin Sill complex today consists of three larger geographically separate sill bodies comes from the shaded relief aeromagnetic anomaly map (Fig. 7). The outcrops of the Whin Sill complex and the related dykes are plotted onto this map and the Permo-Carboniferous dykes match with strong linear magnetic anomalies. In particular, the Holy Island Dyke can be traced further to the east, well offshore. In addition, the aeromagnetic map suggests that the High Green Dyke steps to the north, being in direct connection to the southernmost part of the Alnwick Sill. This further supports the proposal that the High Green Dyke might have formed the feeder to the Alnwick Sill. Even though some of the details of the aeromagnetic anomaly map are lost in conversion into black-and-white reproduction, it is apparent that Whin Sill outcrops bound regions that have the short-wavelength and high-amplitude textures characteristic of a strong and shallow-sourced magnetic anomaly (Fig. 7, location 1). This anomaly is therefore most likely to be the expression of the subsurface continuation of the Whin Sill complex. Onshore the Whin Sill complex is surrounded by relatively smooth magnetic anomalies (Fig. 7, location 2), indicating that these areas are not underlain by the doleritic intrusion. Smooth anomalies also occur between the Holy Island Sill and the Alnwick Sill (Fig. 7, location 3) and arguably between the Alnwick Sill and the Hadrian's Wall–Pennines Sill (Fig. 7, location 4), suggesting that the present-day Whin Sill complex consists of three geographically distinct sill bodies, which are not connected in the subsurface. It is not possible to identify the offshore extension of the three sill areas on the basis of the aeromagnetic anomaly map. The anomaly caused by the dolerite is generally less well defined in the North Sea than it is onshore for two reasons: (1) the flight line separation is greater over the North Sea (3 km compared with 2 km onshore); (2) the thickness of the Permian strata increases rapidly into the North Sea, which smooths the magnetic anomaly signal.

The Holy Island Sill and the Hadrian's Wall–Pennines Sill

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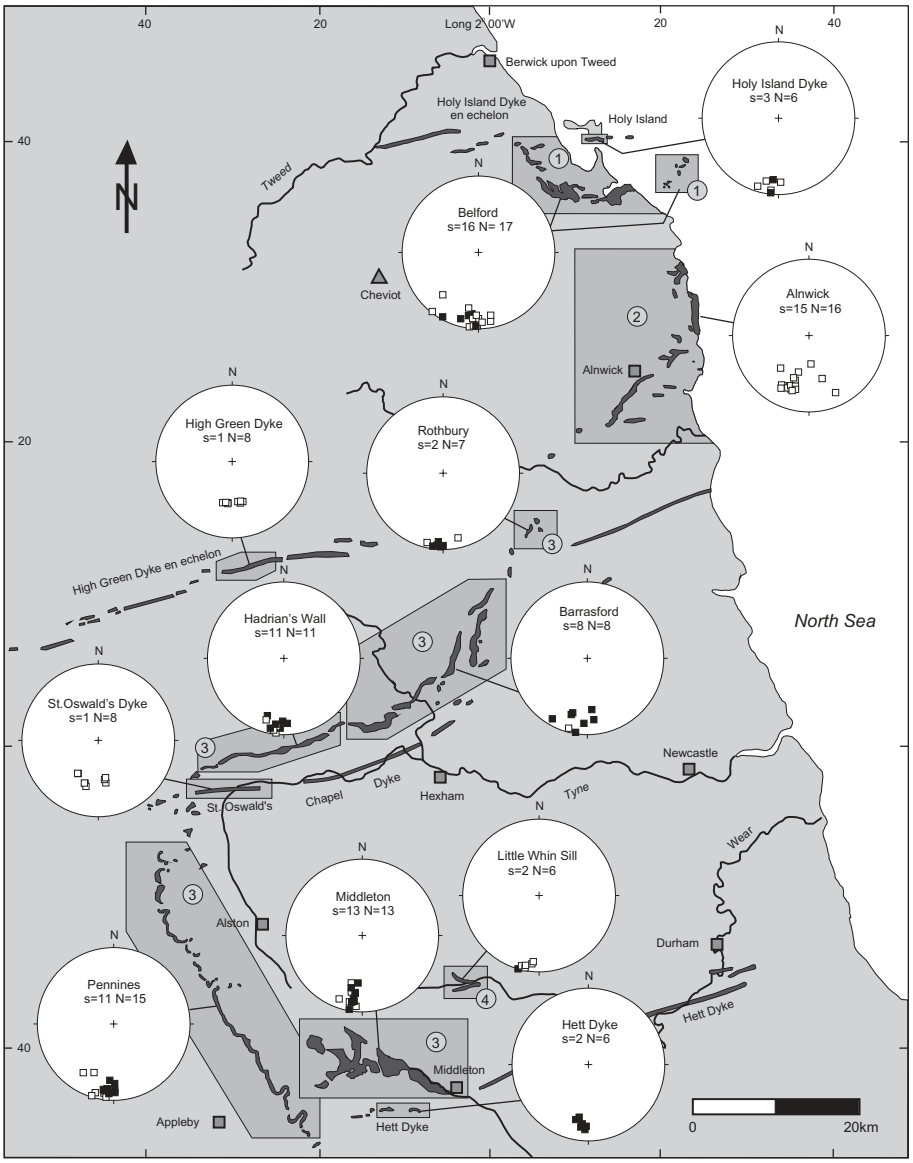


Fig. 6. Map of the study area with the palaeodirectional results plotted for individual regions of the Whin Sill complex and related dykes. The regions for which the data are plotted separately are marked in darker grey. The data are plotted into equal angle stereographic projections; ■, plots into the lower hemisphere of the stereonet; □, plots into the upper hemisphere. Additional information given in the stereonets: name of the area as listed in Tables 3 and 4; s, number of localities; N, number of specimens. The figure reveals that there are two distinctively different palaeodirectional signatures, the first being shallow southerly directed with data plotting into the upper and lower hemispheres of the stereonet and the second being more steeply inclined with data plotting into the upper hemisphere only. On the basis of these palaeodirectional results three geographically separated intrusions can be identified that are labelled 1–3. The Little Whin Sill (4) has been identified as a separate intrusion on petrological evidence (Dunham & Kaye 1965; Harrison 1968).

Table 1. *Mean palaeodirections and VGPs for the individual intrusions of the Whin Sill complex and comparative data from previous studies on the Whin Sill complex*

	p-dec.	p-inc.	No. loc.	No. of spec.	*k*	α_{95}	VGP-E	VGP-S	A95
Holy Island Sill and Holy Island Dyke	189.5	−2.8	20	23	27.6	6.3	346.8	35.4	5.5
Alnwick Sill, High Green and St. Oswald's Chapel Dyke	194.6	−25.6	17	32	20.5	8.1	337.1	47.1	7.7
Hadrian's Wall–Pennines Sill and Hett Dyke	189.2	3.3	47	60	35.0	3.5	347.1	32.9	2.5
Creer *et al*. (1959)	187.8	−4.9	34	99	31.6	4.4	348.9	37.3	
Storetvedt & Gidskehaug (1969)	193.8	−21.7	5	45	259	4.8	339.1	44.4	
Thomas *et al*. (1995)	184.0	−4.4	5	31	38.4	12.5	353	37.1	

p-dec., declination of the mean direction; p-inc., inclination of the mean direction; No. loc, total number of sampling localities that were included in the mean average; No. of spec., total number of specimens that were included in the mean average; *k*, precision parameter of the resulting average; α_{95}, 95% confidence limit for calculated mean direction; VGP-E, VGP longitude; VGP-S, VGP latitude in the southern hemisphere; A95, 95% confidence limit for the VGP.

cannot be separated on the basis of their palaeomagnetic directions, and, as discussed above, both appear to be of similar age. Evidence that the two bodies do represent separate intrusions derives from a study of magma-flow indicators. Macroscopic flow indicators and measurements of anisotropy of magnetic susceptibility (AMS) show that the magma flow in the Holy Island Sill was southward directed and away from its proposed feeder dyke, the Holy Island Dyke (Liss *et al*. 2002). For the Hadrian's Wall–Pennines Sill, in contrast, the magma flow was predominantly northward (Liss 2003), consistent with the Hett Dyke acting as a feeder.

The palaeomagnetism of the Whin Sill complex has been studied previously, notably by Creer *et al*. (1959), Storetvedt & Gidskehaug (1969) and Thomas *et al*. (1995). The early study by

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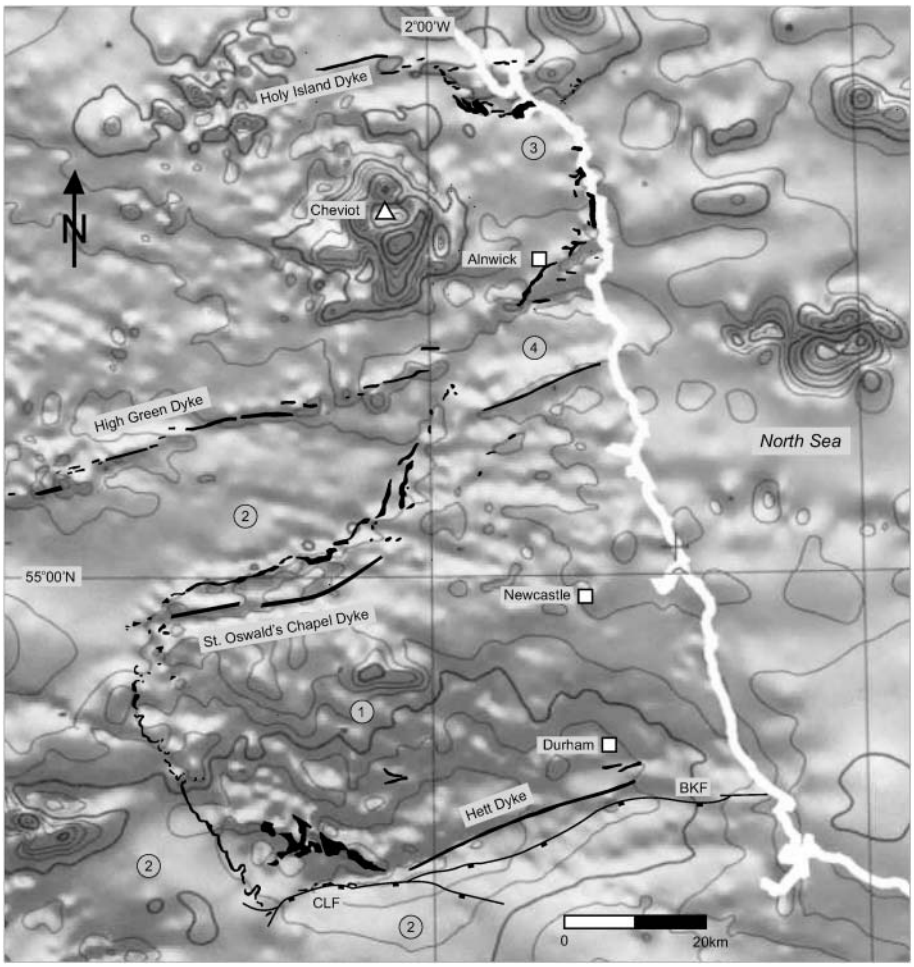


Fig. 7. Aeromagnetic anomaly map of NE England; the thick white line represents the coastline. The surface outcrops of the Whin Sill complex and related dykes are superimposed in black. 1, The subsurface continuation of the Whin Sill complex is expressed by the textured characteristic of a strong and shallow sourced magnetic anomaly. 2, A smooth magnetic anomaly surrounds the Whin Sill complex, indicating that in these areas the intrusion is not present subsurface. 3, 4, Individual regions of the Whin Sill complex are divided by a smooth magnetic signature, demonstrating that the Whin Sill complex consists of three geographically separate intrusions, which are not connected subsurface. The faults restricting the extent of the Whin Sill complex to the south are the Closehouse Lunedale Fault (CLF) and the Butterknowle Fault (BKF). The strong linear WNW–ESE-trending magnetic anomalies are related to Tertiary dykes and are not further discussed. The map is based upon, and converted into grey scales from, the BGS colour shaded relief Magnetic Anomaly Map of Britain, Ireland and adjacent areas, 1:1 500 000 (1998) by permission of the British Geological Survey, IPR/35-34C. Contours are shown at 25 nT intervals; onshore flight lines are north–south with a spacing of 2 km whereas offshore flight lines are east–west with a spacing of 3 km.

Creer *et al*. worked on 36 localities spread along the Whin Sill outcrop (in present terms, Holy Island Sill, seven localities; Alnwick Sill, eight localities; Hadrian's Wall–Pennines Sill, 18 localities (one rejected); Little Whin Sill, three localities (one rejected)). No systematic cleaning was applied; natural remanent magnetization (NRM) directions were scattered in an oval pattern very similar to that of Figure 2. Although Creer *et al*. (1959) calculated a single, tilt-corrected, mean palaeomagnetic direction (declination/inclination (*D*/*I*) 187.8/−4.9) it is now possible to see the non-Fisherian scatter of directions as arising from the mixing of two distinct populations. Near-horizontal field inclinations from the Holy Island Sill and the Hadrian's Wall–Pennines Sill (25 localities) bias the mean, which compares favourably with that found here for these sills (Table 1).

Storetvedt & Gidskehaug (1969) worked on eight localities (Holy Island Sill, two localities; Alnwick Sill, five localities (one unstable); Hadrian's Wall–Pennines Sill, one locality). They applied routine alternating field demagnetization and distinguished two groups of magnetization directions corresponding closely to those recognized here. On the basis of limited thermal demagnetization, however, they rejected all near-horizontal magnetizations as contaminated with a secondary overprint and reported a mean direction of (*D*/*I* 193.8/−21.7), based on five localities, four of which are from the Alnwick Sill (see Table 1). Thomas *et al*. (1995), in a study directed towards palaeointensity determination, reported results from five localities, all in the Hadrian's Wall–Pennines Sill, and supported by extensive rock-magnetic investigation. ChRM directions were isolated by thermal demagnetization; the tilt-corrected mean direction found was (*D*/*I* 180.0/−4.4). Overall, the results from these earlier studies, when placed in their regional contexts, are in agreement with those of the present study.

Virtual geomagnetic poles (VGPs) have been calculated separately for the three areas of the Whin Sill complex and their associated dykes (Table 1) by treating VGPs from individual localities as Fisher distributed. The plot of the mean VGPs and their confidence limits (Fig. 8a) demonstrates the similarity of the pole positions of the Hadrian's Wall–Pennines Sill and the Holy Island Sill. By contrast, there is an angular separation of about 17° between these and the VGP of the Alnwick Sill, a difference that, in terms of the calculated confidence limits, is significant. Figure 8a also shows that the VGPs of Creer *et al*. (1959) and Thomas *et al*. (1995) are close to those of the Hadrian's Wall–Pennines Sill and the Holy Island Sill, whereas the VGP of Storetvedt & Gidskehaug (1969) is close to the position of that for the Alnwick Sill.

A VGP is calculated on the assumption of a geocentric dipole model for the Earth's magnetic field, for an instant reading of the palaeofield direction; non-dipole components of the field are reflected in the scatter of time-equivalent VGP directions. A palaeomagnetic pole, by contrast, is an average, calculated from VGPs (or remanence directions), over a period of time (of the order of 10^5 a) long enough to account for secular variation (see McElhinny & McFadden 2000, p. 22). In palaeomagnetic terms, an igneous intrusion event lasts until all the remanence is blocked in, thus the remanence in a thick sill or dyke may reflect some secular

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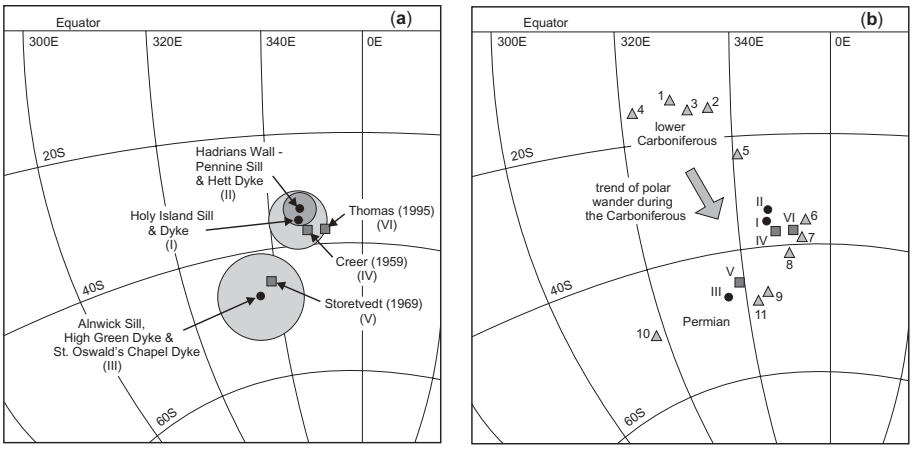


Fig. 8. Plot of the VGPs for the areas of the Whin Sill complex and for a selection of Carboniferous to Permian sites. (**a**) VGPs for the Whin Sill complex; ●, data of this study (I–III) with the corresponding confidence limits (grey circles). The palaeopoles determined by previous studies on the Whin Sill complex (IV–VI) are given by grey squares. Values are listed in Table 1. (**b**) The data of the Whin Sill complex (I–VI) are plotted together with the palaeopoles of selected early Carboniferous to Permian sites (triangles 1–11). The data and references are listed in Tables 1 and 2. Also given is the trend of the apparent polar wander of the British

variation of the geomagnetic field. However, even for a sill of 80 m thickness (the maximum for the Whin Sill complex in outcrop), cooling times are only of the order of 10^2 a and, relatively, instantaneous. On this basis the averaged remanence directions for the three intrusion events identified here are analysed as VGPs. McElhinny & McFadden (2000, p. 79) stated that 'a typical paleomagnetic site should represent, as near as possible, a single point in time relative to the time frame over which the paleomagnetic field changed'. In accordance with this usage, the localities sampled in this study contribute to three palaeomagnetic sites, identified as the three separate sill bodies. There has been a looseness of usage of the term 'site' in the palaeomagnetic literature, resulting in potentially misleading entries to the Global Paleomagnetic Database (see McElhinny & McFadden 2000, p. 230); for example, Storetvedt & Gidskehaug (1969) are reported as measuring 24 samples from five sites. For this reason this paper, following McElhinny & McFadden (2000), distinguishes clearly between the terms (sampling) locality and (palaeomagnetic) site.

In palaeomagnetic terms it makes a significant difference whether the intrusion of the Whin Sill complex is regarded as a single intrusion event (see Storetvedt & Gidskehaug 1969), leading to a single VGP, or, as seen here, as the product of three separate events, characterized by separate VGPs.

The straightforward interpretation of the present data is that the three VGPs represent instances of the palaeofield, with the scatter between them arising from secular variation. In terms of recent behaviour of the geomagnetic field, an angular difference of VGPs of 17° is large but not implausible.

Other interpretations are possible. The present data fall within the Kiaman Reversed superchron (*c*. 311–262 Ma, Opdyke 1995). It has been argued that secular variation is suppressed during superchrons (Courtillot & Besse 1987; McFadden & Merrill 1995), in which case a 17° shift would be highly unusual (and would support the rival view (Kruiver *et al*. 2000, 2002) of non-suppressed secular variation). On the former argument, however, VGPs might be interpreted as pole positions and be compared directly with a relevant apparent polar wander path. In Figure 8b the VGPs from this study are plotted against representative pole positions and VGPs (data in Table 2) from the British Isles, for the Carboniferous–Permian period spanning the time of intrusion of the Whin Sill complex. It is notable that the current results follow this path, whereas secular variation might be expected to cause movement in any direction. The VGPs for the Holy Island Sill and the Hadrian's Wall–Pennines Sill might then be compared with the late Carboniferous–early Permian data (Fig. 8b, 6-8); the radiometric ages reported above, of *c*. 296 Ma, would support this. The VGP for the Alnwick Sill might be compared with the Permian data (Fig. 8b, 9–11), suggesting that the Alnwick Sill represented a later, possibly Permian, intrusion event and a resurrection of activity of the igneous centre. Four of the 21 K–Ar analyses performed by Fitch & Miller (1967) are from the Alnwick Sill but their ages range from 269 ± 18 Ma to

Table 2. *Carboniferous to Permian palaeomagnetic poles for the British Isles*

No.	Formation	Age range	*D*/*I*	α_{95}	VGP-E	VGP-S	Reference
Early Carboniferous							
1	Derbyshire intrusion	Early Carboniferous	211/37	15	329	12	Piper *et al*. 1991
2	Derbyshire lava	*c*. 335 Ma	203/36	8	336	14	Piper *et al*. 1991
3	Kingshorn–Burntisland	Early Carboniferous	205/31	7	332	14	Torsvik *et al*. 1989
4	Largs lava	Early Carboniferous	213/29	8	322	14	Torsvik *et al*. 1989
Early–Late Carboniferous							
5	Dunbar big dyke	Early–Late Carboniferous	196/19	12	341	23	Xu & Tarling 1987
6	Argyll dyke	316–324 Ma		5	355	35	Esang & Piper 1984
Permo-Carboniferous							
7	Midland Valley Sill	Late Carboniferous–Early Permian	182/−9	5	354	38	Torsvik *et al*. 1989
8	Midland Valley Sill	Late Carboniferous–Early Permian	184/5	5	351	41	Thomas *et al*. 1995
Permian							
9	Barra dykes	Early–Late Permian	184/−26	11	346	47	Piper 1992
10	Stornoway	241–256 Ma	024/39	9	318	50	Storetvedt & Steel 1977
11	Exeter lava	Early Permian	189/−19	10	343	48	Cornwell 1967

All poles are reliable in terms of the applied demagnetization procedures.

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316 ± 13 Ma and are thus insufficiently precise to test the suggestion.

Whatever the merits of these arguments, the key contribution of this study is the demonstration, palaeomagnetically, of significant differences within the Whin Sill complex, and the consequent geological implication that the sill is a multi-component body.

Conclusions

The palaeomagnetic analyses of this study have revealed that the dominant carrier of remanent magnetization of the Whin Sill dolerite is relatively pure magnetite in the pseudo-single domain size range. The characteristic remanent magnetization of specimens is typically well defined and unidirectional. The regional palaeodirectional results show distinctive palaeomagnetic signatures. These data in conjunction with an interpretation of the aeromagnetic anomaly map and a combined study on magma-flow indicators and AMS (Liss *et al*. 2002; Liss 2003) allow subdivision of the Whin Sill complex into three larger geographically separate sill bodies. These are named (from north to south) the Holy Island Sill, the Alnwick Sill and the Hadrian's Wall–Pennines Sill. The Little Whin Sill has been identified previously as a fourth, separate sill on the basis of petrological evidence, although palaeomagnetism cannot distinguish it from the surrounding Hadrian's Wall–Pennines Sill. On the basis of the palaeodirectional results it is also possible to suggest possible relationships between the exposed contemporaneous dykes and the individual intrusions. It is proposed that the Holy Island Dyke formed the feeder to the Holy Island Sill, that the High Green Dyke and St. Oswald's Chapel Dyke are contemporaneous with the intrusion of the Alnwick Sill, with the High Green Dyke being a possible feeder to this sill, and that the Hett Dyke probably fed the Hadrian's Wall–Pennines Sill. The study thus demonstrates the potential of palaeomagnetism, in favourable circumstances, to yield hitherto unresolved detail of the development of an igneous intrusion.

Virtual geomagnetic pole positions have been calculated for the three major components of the Whin Sill complex, without application of a tilt correction, which, it is argued, is unnecessary. These are: Holy Island Sill and Holy Island Dyke, 346.8E, 35.4S; Alnwick Sill, High Green Dyke and St. Oswald's Chapel Dyke, 337.1E, 47.1S; Hadrian's Wall–Pennines Sill and Hett Dyke, 347.1E, 32.9S. Although the VGPs for the Holy Island Sill and the Hadrian's Wall–Pennines Sill are very similar, independent evidence indicates that all three VGPs represent distinct igneous events. On this basis the VGP of Storetvedt & Gidskehaug (1969), widely used for the late Carboniferous of the British Isles, should be amended.

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Appendix 1. *Details of the palaeoedirectional results for the individual specimens*

Sample no.	Longitude (W)	Latitude (N)	Area	Group	sus_0	J_i	MAD	p-dec.	p-inc.	VGP-E	VGP-S
WS 3	1°35′50″	55°28′15″	Alnwick	1	46.9	4690	0.4	176.6	−49.6	5.6	65.0
WS 8-211*	1°43′30″	55°37′00″	Belford	1	0.87	48.9	3.4	176.2	−5.7	–	–
WS 8-233*	1°43′30″	55°37′00″	Belford	1	0.84	44	3.3	177.8	−4.5	–	–
WS 8-412	1°43′30″	55°37′00″	Belford	4	5.6	2800	2.0	177.4	−5.3	–	–
WS 9	1°43′30″	55°37′00″	Belford	1	29.1	7900	0.1	169.7	−10.7	–	–
WS 15	1°36′50″	55°31′40″	Alnwick	2	40.8	2068	6.0	200.7	−20.8	330.5	42.6
WS 20	1°37′30″	55°26′45″	Alnwick	3	20.7	3242	2.8	207.5	−17.2	–	–
WS 22	1°37′30″	55°26′45″	Alnwick	1	14.5	2537	2.5	197.9	−27.8	–	–
WS 23	1°38′00″	55°27′30″	Alnwick	1	36.8	5724	1.9	221.4	−31.0	302.6	40.2
WS 24	1°44′40″	55°25′50″	Alnwick	3	5.4	890	6.7	208.4	−14.5	322.3	37.1
WS 26	1°35′50″	55°36′00″	Belford	2	43.2	2700	7.5	209.6	2.0	324.4	28.7
WS 27	1°47′45″	55°36′00″	Belford	1	45.9	2834	1.4	190.6	−17.2	344.2	42.7
WS 29	1°49′35″	55°36′20″	Belford	1	42.9	3688	2.2	182.5	2.4	355.5	33.4
WS 30	1°51′20″	55°37′00″	Belford	1	50.3	4165	1.1	182.9	−11.0	354.7	40.1
WS 32	1°50′10″	55°36′40″	Belford	1	23.8	11720	3.9	181.9	−2.5	356.2	35.9
WS 33	1°52′20″	55°38′20″	Belford	1	1.7	376.2	4.8	195.2	6.4	340.9	30.1
WS 34	1°53′45″	55°38′15″	Belford	1	4.0	919.5	4.6	184.0	−2.2	353.5	35.6
WS 35	1°54′20″	55°38′15″	Belford	3	19.0	6300	5.0	185.6	−8.3	351.3	38.6
WS 36	1°55′50″	55°38′50″	Belford	4	47.6	1449	1.5	189.3	10.7	347.9	28.7
WS 38	1°55′15″	55°37′45″	Belford	4	48.9	1045	2.2	187.7	−1.3	349.0	34.9
WS 43	1°42′45″	55°36′40″	Belford	1	32.2	2128	3.7	180.2	−8.9	358.3	39.1
WS 44	1°44′35″	55°36′40″	Belford	1	42.2	2757	2.8	186.6	12.0	351.1	28.3
WS 48	1°37′00″	55°31′40″	Alnwick	2	76.8	2500	2.9	202.7	−17.0	328.8	40.0
WS 51*	1°37′50″	55°30′30″	Alnwick	1	104.2	2400	18.0	202.1	−25.4	327.6	44.7
WS 59	1°36′00″	55°29′00″	Alnwick	1	43.2	4936	2.5	209.8	−17.6	320.1	38.1
WS 60	1°35′30″	55°29′00″	Alnwick	1	45.9	3900	2.6	163.0	−29.3	23.7	48.3
WS 62	1°36′15″	55°27′40″	Alnwick	1	36.2	3400	3.6	199.4	−17.3	332.8	41.1
WS 63	1°48′15″	55°40′10″	Holy Island Dyke	1	56.5	3191	2.0	191.6	−10.9	–	–
WS 64	1°48′15″	55°40′10″	Holy Island Dyke	4	50.1	1281	3.0	186.0	1.3	–	–
WS 65	1°48′15″	55°40′10″	Holy Island Dyke	1	48.5	1910	4.8	178.2	−10.9	–	–
WS 67	1°46′50″	55°40′10″	Holy Island Dyke	2	88.7	2255	4.5	197.2	−4.1	337.4	34.9

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WS 68	1°47′00″	55°40′10″	Holy Island Dyke	2	73.1	4278	4.0	186.5	−3.9	350.5	36.3
WS 70	1°47′40″	55°40′10″	Holy Island Dyke	2	20.0	446	4.7	185.0	12.0	352.9	28.4
WS74[1]	1°36′20″	55°29′10″	Alnwick	1	123.2	3421	2.0	183.2	−31.5	353.3	51.7
WS 76	1°36′20″	55°29′10″	Alnwick	2	78.6	1397	7.6	197.9	−16.4	334.9	41.0
WS 90	2°26′45″	54°59′40″	Hadrian's Wall	3	2.3	446.5	8.7	196.7	−10.1	336.3	38.3
WS 91	2°26′45″	54°59′40″	Hadrian's Wall	1	46.3	2282	5.0	187.1	8.6	349.5	30.4
WS 93	2°22′55″	55°00′10″	Hadrian's Wall	1	23.9	1819	5.9	191.1	4.9	344.7	31.8
WS 94	2°23′00″	55°00′00″	Hadrian's Wall	1	46.7	2536	7.6	177.9	9.4	0.2	30.2
WS 95	2°20′35″	55°00′40″	Hadrian's Wall	2	29.9	1701	2.1	188.9	−3.5	346.7	36.2
WS 96	2°20′30″	55°00′50″	Hadrian's Wall	1	42.9	3340	2.9	186.2	−6.5	349.9	38.0
WS 98	2°18′40″	55°01′30″	Hadrian's Wall	2	25.9	926	4.7	196.2	14.6	339.9	26.1
WS 99	2°18′50″	55°01′30″	Hadrian's Wall	1	37.7	1298	4.9	183.4	5.3	353.8	32.3
WS 100	2°19′00″	55°01′00″	Hadrian's Wall	2	55.5	2920	3.1	186.8	−1.8	349.4	35.6
WS 101	2°11′45″	55°02′20″	Barrasford	3	36.5	3100	4.9	189.7	1.8	346.2	33.5
WS 103	2°26′35″	54°59′40″	Hadrian's Wall	1	31.9	5307	1.5	181.0	11.3	356.6	29.3
WS 105	2°30′00″	54°59′30″	Hadrian's Wall	2	47.5	961.8	8.3	181.9	−10.2	355.2	40.1
WS 113	2°03′45″	55°07′20″	Barrasford	2	47.6	5094	2.7	210.1	5.9	323.8	26.9
WS 119	2°09′40″	55°03′00″	Barrasford	3	33.2	3000	4.9	195.8	−3.2	338.5	35.0
WS 122	2°07′45″	55°04′30″	Barrasford	1	25.5	2600	4.0	196.4	15.2	349.8	25.7
WS 124	2°02′00″	55°07′20″	Barrasford	1	22.2	4739	1.1	174.9	12.4	3.8	28.5
WS 125	2°01′40″	55°07′40″	Barrasford	1	35.7	4400	4.8	195.5	17.0	341.0	24.9
WS 127	2°01′40″	55°05′30″	Barrasford	2	21.1	3977	2.9	183.6	9.5	353.8	30.1
WS 128	2°03′30″	55°08′00″	Barrasford	1	11.8	2503	2.4	175.7	22.0	2.6	23.4
WS 132	1°54′15″	55°14′30″	Rothbury	1	9.9	4700	1.8	188.2	2.2	–	–
WS 133	1°54′15″	55°14′30″	Rothbury	1	9.9	6700	1.2	167.6	−8.2	–	–
WS 134	1°54′15″	55°14′30″	Rothbury	1	15.9	5320	1.4	180.8	3.5	–	–
WS 135	1°52′40″	55°15′45″	Rothbury	1	57.7	2300	4.5	193.9	−4.0	–	–
WS 136-6	1°52′40″	55°15′45″	Rothbury	1	58.7	2500	2.2	184.8	6.0	–	–
WS 136-3	1°52′40″	55°15′45″	Rothbury	1	57.3	2270	3.2	182.4	2.0	–	–
WS 137	1°52′40″	55°15′45″	Rothbury	1	45.6	2360	2.6	185.5	3.7	–	–
WS 143	1°35′40″	55°27′40″	Alnwick	1	26.0	4807	2.4	196.9	−37.0	331.7	53.1
WS 146	1°37′50″	55°26′15″	Alnwick	1	31.7	3187	4.1	195.9	−21.4	336.6	44.1
WS 147	1°38′30″	55°25′45″	Alnwick	2	7.6	1215	5.1	194.8	−18.5	338.5	42.8
WS 149	1°38′45″	55°25′15″	Alnwick	1	47.2	3054	3.4	155.9	−11.3	29.1	36.8
WS 150	1°45′45″	55°21′45″	Alnwick	1	43.3	2851	2.1	200.9	−29.1	328.2	47.2
WS 151-12	2°28′30″	54°57′35″	St. Oswald's Dyke	3	94.4	1721	6.3	211.5	−36.7	–	–
WS 151-2	2°28′30″	54°57′35″	St. Oswald's Dyke	2	94.6	1673	8.9	198.8	−29.4	–	–
WS 151-3	2°28′30″	54°57′35″	St. Oswald's Dyke	2	92.2	1560	12.3	212.8	−36.1	–	–
WS 152-1	2°28′30″	54°57′35″	St. Oswald's Dyke	4	71.7	572	9.8	195.7	−26.0	–	–
WS 152-3	2°28′30″	54°57′35″	St. Oswald's Dyke	1	75.3	815	12.9	197.2	−29.5	–	–
WS 153-2	2°28′30″	54°57′35″	St. Oswald's Dyke	3	91.0	1254	3.9	170.1	−31.8	–	–
WS 153-4	2°28′30″	54°57′35″	St. Oswald's Dyke	1	88.5	1075	8.8	169.2	−37.9	–	–
WS 153-9	2°28′30″	54°57′35″	St. Oswald's Dyke	1	87.7	1140	9.5	171.9	−35.1	–	–
WS 155	2°34′30″	54°54′55″	Pennines	1	50.8	3175	5.5	197.9	−1.5	335.8	34.2
WS 156	2°37′30″	54°52′25″	Pennines	1	19.9	1609	2.3	195.1	−4.5	338.8	36.3
WS 157	2°37′40″	54°51′20″	Pennines	1	13.8	1943	2.3	179.7	8.2	–	–
WS 158	2°37′40″	54°51′20″	Pennines	1	54.7	2239	5.4	179.9	6.8	–	–
WS 159	2°37′00″	54°49′30″	Pennines	1	38.6	4122	4.5	184.4	5.6	352.4	32.6
WS 160	2°38′15″	54°49′30″	Pennines	1	32.2	5623	2.0	179.4	14.6	358.3	28.1
WS 162	2°38′15″	54°49′30″	Pennines	2	28.4	1915	3.0	212.7	−16.0	315.7	36.8
WS 163	2°37′35″	54°48′50″	Pennines	4	67.0	2811	1.5	180.2	−9.9	–	–
WS 164	2°37′35″	54°48′50″	Pennines	1	70.9	3386	3.8	186.0	−2.2	–	–
WS 166	2°34′50″	54°42′45″	Pennines	2	64.3	2142	1.5	202.9	−21.2	326.2	42.9
WS 170	2°12′00″	54°38′50″	Middleton	4	45.4	1631	3.0	192.8	20.7	344.1	24.0
WS 171	2°24′00″	54°37′45″	Pennines	1	20.9	2252	1.7	189.5	8.3	–	–
WS 172	2°24′00″	54°37′45″	Pennines	1	33.0	2296	4.2	186.5	9.5	–	–
WS 173	2°26′35″	54°38′45″	Pennines	1	20.9	3251	2.6	184.6	17.1	–	–
WS 174	2°26′35″	54°38′45″	Pennines	1	33.4	4206	1.7	188.4	7.7	–	–
WS 176	2°22′15″	54°36′10″	Pennines	1	57.9	3304	6.8	188.4	−5.9	347.1	38.1
WS 177	2°20′30″	54°38′00″	Middleton	2	9.0	980.7	9.9	189.8	7.5	346.3	31.3
WS 179	2°13′50″	54°38′50″	Middleton	1	1.5	68.2	5.3	187.2	8.0	349.4	31.3
WS 181	2°12′55″	54°39′10″	Middleton	1	39.7	2786	3.7	187.5	16.2	349.5	27.0
WS 182	2°09′00″	54°38′45″	Middleton	1	33.7	1298	5.7	193.3	−25.8	338.4	47.9
WS 183	2°08′40″	54°38′30″	Middleton	1	20.9	1900	8.6	188.8	10.3	347.8	30.0
WS 184	2°11′20″	54°39′00″	Middleton	1	46.7	2131	4.7	191.4	−7.4	343.3	38.5
WS 185	2°11′25″	54°39′20″	Middleton	1	44.0	3100	3.3	187.8	15.5	349.2	27.4
WS 186	2°11′25″	54°39′20″	Middleton	4	41.5	1117	5.4	191.6	−16.4	342.1	43.1
WS 190-1	2°12′10″	54°36′00″	Hett Dyke	1	51.2	3557	4.1	193.0	17.3	–	–
WS 190-5	2°12′10″	54°36′00″	Hett Dyke	1	52.7	3910	5.0	190.6	20.3	–	–
WS 191-1	2°06′10″	54°36′25″	Hett Dyke	1	21.5	1930	6.6	186.4	14.9	–	–
WS 191-5	2°06′10″	54°36′25″	Hett Dyke	1	19.6	1920	3.2	187.6	11.9	–	–
WS 192-8	2°06′10″	54°36′25″	Hett Dyke	1	0.9	27	2.2	182.6	12.6	–	–
WS 192-2	2°06′10″	54°36′25″	Hett Dyke	1	0.9	25	2.4	183.9	9.4	–	–
WS 195	2°17′15″	54°39′25″	Middleton	4	40.5	874.7	8.6	200.2	−7.3	332.4	36.7

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WS 197	2°09′20″	54°38′20″	Middleton	1	42.0	2697	1.6	185.7	−4.1	350.7	37.5
WS 198	2°07′00″	54°37′25″	Middleton	1	44.2	3217	4.4	190.8	1.9	344.9	34.0
WS 199	2°07′10″	54°37′35″	Middleton	1	16.1	1559	7.3	185.2	26.4	352.5	21.5
WS 202-0	2°01′45″	54°44′45″	Little Whin Sill	1	45.5	4724	1.9	191.1	−4.0	–	–
WS 202-11	2°01′45″	54°44′45″	Little Whin Sill	1	45.5	4150	1.5	191.2	−2.6	–	–
WS 203-1	2°01′45″	54°44′45″	Little Whin Sill	1	42.9	4458	2.2	194.7	−3.4	–	–
WS 203-3	2°01′45″	54°44′45″	Little Whin Sill	1	44.9	4724	4.9	196.1	0.1	–	–
WS 204-32	2°01′45″	54°44′45″	Little Whin Sill	1	44.1	6140	3.4	185.6	−8.5	–	–
WS 204-31	2°01′45″	54°44′45″	Little Whin Sill	1	43.9	6188	4.6	186.4	−6.7	–	–
WS 215-11	2°30′40″	55°10′25″	High Green Dyke	1	54.6	925	3.1	168.7	−34.9	–	–
WS 215-21	2°30′40″	55°10′25″	High Green Dyke	1	52.2	909	1.5	172.8	−34.6	–	–
WS 215-32	2°30′40″	55°10′25″	High Green Dyke	1	57.0	935	2.3	169.5	−32.5	–	–
WS 215-42	2°30′40″	55°10′25″	High Green Dyke	1	45.9	854	2.6	165.5	−33.2	–	–
WS 216-11	2°30′40″	55°10′25″	High Green Dyke	1	8.5	939	0.9	189.0	−33.6	–	–
WS 216-21	2°30′40″	55°10′25″	High Green Dyke	1	5.6	1039	1.0	193.8	−32.1	–	–
WS 216-31	2°30′40″	55°10′25″	High Green Dyke	1	12.1	914	1.0	186.1	−32.6	–	–
WS 216-42	2°30′40″	55°10′25″	High Green Dyke	1	14.8	841	1.2	186.9	−33.0	–	–
WS 226-21	1°36′15″	55°38′25″	Belford	1	43.1	1696	4.0	170.6	−5.5	10.4	36.8
WS 229-11	1°37′20″	55°38′00″	Belford	2	41.5	648	10.2	220.2	−18.2	308.5	34.3
WS 231-14	1°39′20″	55°37′00″	Belford	2	43.5	684	3.9	218.5	−1.3	314.7	27.0

Group indicates demagnetization behaviour as subdivided in Figure 3; sus_0, room temperature susceptibility (10^{-3}); J_i, remanence intensity at room temperature (mA m^{-1}); MAD, mean angular deviation of the isolated ChRM; p-dec., declination of the ChRM component; p-inc., inclination of the ChRM component; VGP-E, longitude (E) of the virtual geomagnetic pole; VGP-S, latitude (S) of the virtual geomagnetic pole. VGPs were calculated only for localities that were not subject to further block or site averaging (see Appendix 2).
*Samples with atypical magnetic behaviour; their results were rejected from further analyses.

Appendix 2. *Details of the block averages for the palaeodirectional results*

	Longitude (W)	Latitude (N)	Area	No. of spec.	k	α_{95}	p-dec.	p-inc.	VGP-E	VGP-S
Block averages										
8+9	1°43′30″	55°37′00″	Belford	2	296.9	5.3	175.3	−6.6	4.5	37.8
136	1°52′40″	55°15′45″	Rothbury	2	621.2	10.0	183.6	4.0	–	–
151	2°29′20″	54°57′35″	St. Oswald's Dyke	3	112.8	11.7	207.5	−34.3	–	–
152	2°29′20″	54°57′35″	St. Oswald's Dyke	2	939.6	8.2	196.5	−27.8	–	–
153	2°29′20″	54°57′35″	St. Oswald's Dyke	3	616.0	5.0	170.4	−34.9	–	–
190	2°12′10″	54°36′00″	Hett Dyke	2	944.2	8.1	191.8	18.8	345.0	25.2
191	2°06′10″	54°36′25″	Hett Dyke	2	1320.0	6.9	187.0	13.4	–	–
192	2°06′10″	54°36′25″	Hett Dyke	2	1127.0	7.4	183.3	11.0	–	–
202	2°01′45″	54°44′45″	Little Whin Sill	2	6039.0	3.2	191.1	−3.3	–	–
203	2°01′45″	54°44′45″	Little Whin Sill	2	894.0	8.4	195.4	−1.7	–	–
204	2°01′45″	54°44′45″	Little Whin Sill	2	3167.0	4.4	186.0	−7.6	350.7	39.1
215	2°30′40″	55°10′25″	High Green Dyke	4	1442.0	2.4	170.1	−34.1	–	–
216	2°30′40″	55°10′25″	High Green Dyke	4	722.8	3.4	188.9	−32.8	–	–
Block averages 2										
20+22	1°37′30″	55°26′45″	Alnwick	2	68.4	30.7	202.9	−22.6	327.3	42.9
63+64+65	1°48′15″	55°40′10″	Holy Island Dyke	3	70.6	14.8	185.3	−6.8	351.8	37.8
132+133+134	1°54′15″	55°14′30″	Rothbury	3	44.0	18.8	178.9	−0.8	359.8	35.2
135+136+137	1°52′40″	55°15′45″	Rothbury	4	130.0	10.9	187.7	1.2	350.5	33.6
151+152+153	2°29′20″	54°57′35″	St. Oswald's Dyke	8	24.8	25.3	191.7	−33.3	339.9	52.6
157+158	2°37′40″	54°51′20″	Pennines	2	6151.0	3.2	179.8	7.5	357.9	31.7
163+164	2°37′35″	54°48′50″	Pennines	2	140.4	21.2	183.1	−6.0	353.7	38.5
171+172	2°24′00″	54°37′45″	Pennines	2	1274.7	7.0	188.0	8.9	348.5	30.8
173+174	2°26′35″	54°38′45″	Pennines	2	130.4	22.0	186.5	12.4	350.3	29.1
190	2°12′10″	54°36′00″	Hett Dyke	2	944.2	8.1	191.8	18.8	345.0	25.2
191+192	2°06′10″	54°36′25″	Hett Dyke	4	697.0	9.5	185.1	12.2	352.1	29.3
202+203	2°01′45″	54°44′45″	Little WS	4	625.0	10.0	193.3	−2.5	341.5	36.0
215+216	2°30′40″	55°10′25″	High Green Dyke	8	53.2	34.9	179.6	−33.8	357.1	53.4

If more than one specimen was analysed per block, vector averages were determined. The first part of the table lists the average results for specimens that were taken from one block, and the second part averages the former results from outcrop localities where more than one block were taken. No. of spec., total number of specimens that were included in a block or site average; k, precision parameter of the resulting average; α_{95}, 95% confidence limit for calculated mean direction; p-dec., declination of the mean direction; p-inc., inclination of the mean direction; VGP-E, longitude (E) of the virtual geomagnetic pole; VGP-S, latitude (S) of the virtual geomagnetic pole. For block mean directions, VGPs were calculated only if the results were not subject to further site averaging.

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